

A Möbius-invariant Green-circle endpoint norm for the radial dilation symmetry

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Abstract

Adimurthi and Tintarev ask for a sharp critical nonlinearity on $H_0^1(\mathbb{B})$ invariant under the product symmetry suggested by Möbius shifts and the radial dilations $h_s u(r) = s^{-1/2} u(r^s)$. We give a positive answer in the exact sense in which the source defines these actions. A maximal normalized L^2 -mean over all Green-function level circles defines a norm $\|\cdot\|_{\mathcal{G}}$ on $H_0^1(\mathbb{B})$. It is invariant under every disk automorphism and satisfies the sharp inequality

$$\|u\|_{\mathcal{G}} \leq \frac{1}{\sqrt{2\pi}} \|\nabla u\|_2.$$

On radial functions it is exactly $\sup_{0 < r < 1} |u(r)| / \sqrt{\log(1/r)}$, the source's sharp dilation-invariant endpoint. We also show that no nonzero pointwise integral against hyperbolic area can possess both symmetries, explaining why the common endpoint must be nonlocal.

1 The source question

Let $\mathbb{B} = \{z \in \mathbb{C} : |z| < 1\}$. The source defines the Möbius maps

$$\eta_a(z) = \frac{z - a}{1 - \bar{a}z}$$

on $H_0^1(\mathbb{B})$ and, only on the radial subspace $H_{0,r}^1(\mathbb{B})$, the group

$$h_s u(r) = s^{-1/2} u(r^s), \quad s > 0. \tag{1}$$

It identifies the sharp radial endpoint

$$\|u\|_{\infty, \log} = \sup_{0 < r < 1} \frac{|u(r)|}{\sqrt{\log(1/r)}} \tag{2}$$

and asks:

It remains an open problem, however, to find a sharp critical nonlinearity that is invariant with respect to the product group.

The exact source page, including (1) and (2), is reproduced below.

The Trudinger-Moser functional $\int_B e^{4\pi u^2} dx$ fails to be invariant with respect to either of these groups. This, however, happens to testify not for irrelevance of these groups but for an observation that the Trudinger-Moser functional is not the sharp critical nonlinearity and can be replaced by a stronger expression. It remains an open problem, however, to find a sharp critical nonlinearity that is invariant with respect to the product group. The details are as follows.

1.1 Dilation-invariant nonlinearity

In this paragraph we summarize results of [3]. Let $H_{0,r}^1(B)$ denote the subspace of radial functions of $H_0^1(B)$. The transformations

$$h_s u(r) \stackrel{\text{def}}{=} s^{-\frac{1}{2}} u(r^s), \quad u \in H_{0,r}^1(B) \quad s > 0, \quad (1.4)$$

preserve the norm $\|\nabla u\|_2$ of $H_{0,r}^1(B)$, as well as the 2-dimensional Hardy functional $\int_B \frac{u^2}{|x|^2(\log 1/|x|)^2} dx$ (for the Hardy inequality in dimension 2 see Adimurthi and Sandeep [2]). Furthermore, these transformation preserve the norms of a family of weighted L^p -spaces, $p = [2, \infty]$, analogous to the weighted- L^p scale with $p \in [2, 2^*]$ produced by Hölder inequality in the case $N > 2$, interpolating between the Hardy term $\int \frac{u^2}{|x|^2} dx$ and the critical nonlinearity $\int |u|^{2^*} dx$. In the case $N = 2$, the critical exponent is formally $2^* = +\infty$ and the dilation-invariant L^{2^*} -norm is

$$\|u\|_\infty = \sup_{r \in (0,1)} \frac{|u(r)|}{(\log \frac{1}{r})^{1/2}}. \quad (1.5)$$

The Trudinger-Moser nonlinearity is not, however, dilation-invariant. On the other hand it is continuous with respect to the norm (1.5), which means that the L^∞ -nonlinearity (1.5) gives a sharp, dilation-invariant improvement of the Trudinger-Moser nonlinearity, even if only for the subspace of radial functions.

Because (1) is defined only for radial functions, the literal common-extension problem is to find a Möbius-invariant quantity on all of $H_0^1(\mathbb{B})$ whose radial restriction is (2). We solve precisely that problem.

2 Definition and intuition

Write

$$\phi_a(w) = \eta_a^{-1}(w) = \frac{a + w}{1 + \bar{a}w}.$$

The curve $\phi_a(|w| = e^{-t})$ is the level set

$$\log \frac{1}{|\eta_a(z)|} = t$$

of the Green function of the disk with pole at a . Uniform angular measure in the w -coordinate is harmonic measure on this circle seen from its center.

For $u \in C_c^\infty(\mathbb{B})$ define

$$\|u\|_{\mathcal{G}} = \sup_{\substack{a \in \mathbb{B} \\ t > 0}} \left(\frac{1}{2\pi t} \int_0^{2\pi} \left| u\left(\phi_a(e^{-t+i\theta})\right) \right|^2 d\theta \right)^{1/2}. \quad (3)$$

The factor $t^{-1/2}$ is forced by the one-dimensional radial Cauchy–Schwarz inequality and is exactly the scaling in (2). The supremum over centers is what restores Möbius invariance.

3 The full endpoint theorem

Theorem 1. *Formula (3) extends to a norm on $H_0^1(\mathbb{B})$ with the following properties.*

1. For every $u \in H_0^1(\mathbb{B})$,

$$\|u\|_{\mathcal{G}} \leq \frac{1}{\sqrt{2\pi}} \|\nabla u\|_{L^2(\mathbb{B})}, \quad (4)$$

and the constant $1/\sqrt{2\pi}$ is optimal.

2. For every $T \in \text{Aut}(\mathbb{B})$,

$$\|u \circ T\|_{\mathcal{G}} = \|u\|_{\mathcal{G}}. \quad (5)$$

3. If u is radial, then

$$\|u\|_{\mathcal{G}} = \sup_{0 < r < 1} \frac{|u(r)|}{\sqrt{\log(1/r)}}. \quad (6)$$

Consequently $\|h_s u\|_{\mathcal{G}} = \|u\|_{\mathcal{G}}$ for every $s > 0$.

4. More generally, on the subspace of functions radial about any $a \in \mathbb{B}$, the norm is invariant under the conjugate dilation centered at a .

Proof. Fix $a \in \mathbb{B}$ and set $v = u \circ \phi_a$. For $r = e^{-t}$, the zero boundary trace and Cauchy–Schwarz on almost every radial line give

$$|v(r, \theta)|^2 = \left| \int_r^1 \partial_\rho v(\rho, \theta) d\rho \right|^2 \leq \left(\int_r^1 |\partial_\rho v|^2 \rho d\rho \right) \left(\int_r^1 \frac{d\rho}{\rho} \right).$$

The last factor is t . Average in θ , divide by $2\pi t$, and use conformal invariance of Dirichlet energy:

$$\frac{1}{2\pi t} \int_0^{2\pi} |v(e^{-t}, \theta)|^2 d\theta \leq \frac{1}{2\pi} \int_{\mathbb{B}} |\nabla v|^2 = \frac{1}{2\pi} \int_{\mathbb{B}} |\nabla u|^2.$$

Taking the supremum proves (4). Minkowski’s inequality on every circle gives the triangle inequality, and the bound makes (3) continuous in the Dirichlet norm, hence it extends from $C_c^\infty(\mathbb{B})$ to $H_0^1(\mathbb{B})$. The trace formula holds for the extension with an essential supremum. If the extended value is zero, the terms with $a = 0$ show that u vanishes on almost every centered circle, and hence $u = 0$ almost everywhere. Thus it is a norm.

For Möbius invariance, take $T \in \text{Aut}(\mathbb{B})$. Both $T \circ \phi_a$ and $\phi_{T(a)}$ map 0 to $T(a)$. Therefore, for some rotation $R_\gamma(w) = e^{i\gamma}w$,

$$T \circ \phi_a = \phi_{T(a)} \circ R_\gamma. \quad (7)$$

Rotation preserves angular measure. Apply (7) inside the circle mean and use that $a \mapsto T(a)$ is a bijection of $\mathbb{B}$. This proves (5).

Now let u be radial and put

$$M = \sup_{0 < r < 1} \frac{|u(r)|}{\sqrt{\log(1/r)}}.$$

The center $a = 0$ in (3) gives $\|u\|_{\mathcal{G}} \geq M$. Conversely, $|u(z)|^2 \leq M^2 \log(1/|z|)$. Jensen's formula yields, for $0 < r < 1$,

$$\frac{1}{2\pi} \int_0^{2\pi} -\log |\phi_a(re^{i\theta})| d\theta = -\log \max\{|a|, r\} \leq -\log r. \quad (8)$$

Indeed, the mean of $\log |a + re^{i\theta}|$ is $\log \max\{|a|, r\}$, while the mean of $\log |1 + \bar{a}re^{i\theta}|$ is zero. Substituting the pointwise bound into every circle average and using (8) proves the reverse inequality $\|u\|_{\mathcal{G}} \leq M$, hence (6).

If $q = r^s$, direct substitution in (2) gives

$$\sup_r \frac{s^{-1/2}|u(r^s)|}{\sqrt{\log(1/r)}} = \sup_q \frac{|u(q)|}{\sqrt{\log(1/q)}}.$$

This proves radial dilation invariance. Finally, a function radial about a becomes radial about 0 after composition with ϕ_a . Conjugate the preceding identity and use (5).

It remains to prove sharpness. For $T > 0$ define the Moser hat

$$m_T(r) = \frac{\min\{\log(1/r), T\}}{\sqrt{2\pi T}}. \quad (9)$$

In the variable $t = \log(1/r)$,

$$\|\nabla m_T\|_2^2 = 2\pi \int_0^\infty |m'_T(t)|^2 dt = 1.$$

Formula (6) gives $\|m_T\|_{\mathcal{G}} = 1/\sqrt{2\pi}$, proving that the constant in (4) is exact. $\square$

Remark 1 (Endpoint noncompactness). *The hats satisfy $h_s m_T = m_{T/s}$. As $T \rightarrow \infty$ they converge weakly to zero in $H_0^1(\mathbb{B})$, while both their Dirichlet norm and $\mathcal{G}$ -norm remain fixed. Thus the embedding (4) is noncompact along exactly the source dilation symmetry. Möbius shifts move the same concentration to an arbitrary center.*

4 Why a pointwise integral cannot work

The source's Möbius-invariant Trudinger–Moser functional is a local integral against hyperbolic area

$$d\mu(z) = \frac{dx dy}{(1 - |z|^2)^2}.$$

The next proposition shows that no nonzero functional in this natural local class can also be dilation invariant.

Proposition 1. *Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $F(0) = 0$, and set*

$$J_F(u) = \int_{\mathbb{B}} F(u) d\mu.$$

If, for some fixed $s \neq 1$,

$$J_F(h_s u) = J_F(u)$$

for every radial $u \in C_c^\infty(\mathbb{B})$, then $F \equiv 0$.

Proof. In polar coordinates,

$$J_F(u) = 2\pi \int_0^1 F(u(r)) \frac{r}{(1-r^2)^2} dr.$$

In $J_F(h_s u)$ make the change of variable $q = r^s$. This gives

$$J_F(h_s u) = 2\pi \int_0^1 F(s^{-1/2} u(q)) \frac{1}{s} \frac{q^{2/s-1}}{(1-q^{2/s})^2} dq.$$

Use smooth radial plateaus converging to an arbitrary constant c on an arbitrary compact subinterval of $(0, 1)$, then apply Lebesgue differentiation. Invariance forces

$$F(s^{-1/2} c) \frac{1}{s} \frac{q^{2/s-1}}{(1-q^{2/s})^2} = F(c) \frac{q}{(1-q^2)^2} \quad (10)$$

for almost every $q \in (0, 1)$. If $F(c) \neq 0$, then (10) says that

$$s q^{2-2/s} \frac{(1-q^{2/s})^2}{(1-q^2)^2}$$

is constant in q . It is not constant when $s \neq 1$ (compare its limits at $q = 0$ and $q = 1$). Hence $F(c) = 0$. Since c was arbitrary, $F \equiv 0$. $\square$

The nonlocal Green-circle average in (3) is therefore not an ornament: within the natural pointwise hyperbolic class, nonlocality is forced.

5 Verification, novelty, and limitations

Independent checks. The script `verify_green_circle_norm.py` performs deterministic quadrature and complex-algebra checks of:

- the Jensen identity (8) on off-center circles;
- the factorization (7) pointwise;
- the radial endpoint bound on sampled Moser-type profiles;
- the dilation scaling and the exact constant $1/\sqrt{2\pi}$;
- the nonconstant radial-weight ratio in Proposition 1.

The proof itself is exact and does not depend on the numerical checks.

Literature audit. Exact-phrase searches, the run's cheap indexes, searches for Möbius-invariant circular-mean Sobolev norms, and inspection of the related later papers [2, 3, 4] did not locate (3) or a prior resolution of this question. This is evidence, not a definitive novelty certification.

Interpretation and limitation. The result is a full positive answer under the source's own terminology: the source calls the radial norm (2) an L^∞ -nonlinearity, and (3) extends it exactly. The source dilation is not defined on arbitrary nonradial functions, so no claim of such an action is made. Instead, the result supplies full Möbius invariance and dilation invariance on every centered radial subspace, which is the strongest literal common action available. The result does not assert that $\|\cdot\|_{\mathcal{G}}$ is a pointwise integral; Proposition 1 rules out that more restrictive form.

Human review recommendation. Check the trace extension, (8), and the intended meaning of product group. Search the conformally invariant Dirichlet-space and maximal-function literature for the Green-circle norm. If both the proof and novelty survive, the source authors should be contacted.

References

- [1] Adimurthi and K. Tintarev, *On a version of Trudinger–Moser inequality with Möbius shift invariance*, arXiv:0909.0692.
- [2] Adimurthi and K. Tintarev, *On compactness in the Trudinger–Moser inequality*, arXiv:1110.3647.
- [3] C. Tintarev, *Trudinger–Moser inequality with remainder terms*, arXiv:1204.5332.
- [4] Y. Yang and X. Zhu, *Trudinger–Moser embedding on the hyperbolic space*, arXiv:1306.0822.